

EE 5193 — FPGA and HDL

Summer 2026

Electrical and Computer Engineering Department

University of Texas at San Antonio

A Multi-Cycle 16-Bit RISC Processor on the Nexys A7

Course Project Report

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July 10, 2026

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2. Problem Statement

The goal of this project is to design, simulate, and implement a small Reduced Instruction Set Computer (RISC) on the Digilent Nexys A7 board (Artix-7 XC7A100T-1CSG324C) using Vivado. The processor executes 16-bit instructions whose leftmost four bits are the opcode, and every functional block in the datapath/controller diagram has to be written as its own Verilog module.

The controller is specified to have exactly two inputs — the 16-bit IR value and a flag RF_Rp_zero from the register file indicating that the value on read port P is zero — and it must sequence every instruction through the same four steps:

1. **Fetch** — read the instruction at memory location PC into the IR,
2. **Decode** — interpret the opcode and register/address fields,
3. **Execute** — perform the operation,
4. **Update PC** — point the PC at the next memory location.

Memory locations 201 and 202 hold initial operand values. The demonstration program below loads them, computes with them, and stores three results at locations 203, 204, and 205, which must appear on the board's 7-segment display once the program has finished:

#	Mnemonic	Encoding
0	LW r5, 201	1001_0101_1100_1001
1	LW r6, 202	1001_0110_1100_1010
2	ADD r7, r5, r6	0000_0111_0101_0110
3	SW r7, 203	1010_0111_1100_1011
4	LI r8, 250	1000_1000_1111_1010
5	SUB r4, r8, r5	0001_0100_1000_0101
6	SW r4, 204	1010_0100_1100_1100
7	SRA r3, r7	0110_0011_0111_0000 *
8	XOR r2, r3, r4	0100_0010_0011_0100
9	SW r2, 205	1010_0010_1100_1101

* The handout gives 0110_0010_0111_0000 for this instruction, whose destination field decodes to r2, not r3. That contradicts both the mnemonic and the following XOR, which reads r3. I use the corrected encoding here; Section 10.1 describes how I found the discrepancy in simulation.

The deliverables are the Verilog modules, a block diagram, ASM-D charts, a controller output table per opcode, simulation results, and this report.

3. Approach

3.1 Multi-cycle organization

I implemented the processor as a multi-cycle machine: one controller FSM state per architectural step, so every instruction takes exactly four states (Fetch, Decode, Execute, UpdatePC). This maps one-to-one onto the four-step process the assignment requires, which makes the controller easy to specify, easy to draw as an ASM-D chart, and easy to check in a waveform — the state register tells you exactly which

step you are looking at. A single-cycle design would have needed separate instruction and data memories (the assignment wants one memory holding both program and data), and a pipelined design is unjustifiable for a ten-instruction demo.

The FSM has a fifth, terminal state, **Halt**. The demo program is ten instructions long and the assignment says results are displayed “once all instructions have executed,” so the machine needs a defined way to stop. I assigned opcode 1111 to a HALT instruction, placed one at address 10 (right after the program), and made the controller park in the Halt state with a done output high. done gates the display and drives LED0, and the testbench uses it to know when to check results.

3.2 One unified memory

Program and data share a single 256×16 -bit memory. Addresses 0–10 hold the program (plus HALT), 201/202 hold the input operands, and 203–205 receive the results. A unified memory means one address bus and a single two-way address mux: during Fetch the address is the PC, and during the Execute step of LW/SW it is the 8-bit address field IR[7:0]. Since the machine is multi-cycle, instruction fetch and data access never happen in the same state, so one port is enough and there is no structural hazard.

The CPU port of the memory uses a **combinational read** with a synchronous write. This is a deliberate choice: with an asynchronous read, the fetched word is available at the memory output in the same state the address is applied, so Fetch and the LW Execute step each fit in a single FSM state and the four-step model stays exact. Vivado infers distributed (LUT) RAM for this style, which costs a few hundred LUTs at 256×16 — cheap on an XC7A100T. My first version used a registered read that maps to block RAM, and the one-cycle read latency quietly broke the four-step model; Section 10.2 covers that bug and why I settled on distributed RAM.

Initial contents (the ten instructions, the HALT, and the operands 25 and 35 at 201/202) are loaded from program.mem with \$readmemb, so the same image feeds both simulation and synthesis.

3.3 Controller

The controller matches the assignment's interface exactly: its only inputs (besides clock, reset, and the clock enable) are IR[15:0] and RF_Rp_zero. It is a five-state FSM (the four steps plus Halt). All outputs are generated in a single combinational block with default assignments first, so no latches are inferred; the opcode IR[15:12] selects the Execute-step outputs, and RF_Rp_zero is consulted in exactly one place.

About that flag: none of the ten demo instructions is conditional, but the assignment specifies RF_Rp_zero as a controller input, so I gave it a real job rather than leaving a dangling port. I defined a conditional branch, JZ rs, addr (opcode 1100, same field layout as SW): if the register on read port P is zero, the PC loads IR[7:0] in the UpdatePC step instead of incrementing. The demo program never executes a JZ, but the path is implemented, simulated, and documented in the controller table, and it is the reason the PC has a load input as well as an increment.

3.4 Reuse of previously developed modules

Two blocks in this design are adaptations of modules I built earlier in the course rather than new work:

- **ALU/shifter** (`alu.v`): the arithmetic unit from the earlier combinational-design lab, trimmed to the four operations this ISA needs (ADD, SUB, XOR, and a one-bit arithmetic right shift). The shift is the sign-replicating $\{a[15], a[15:1]\}$ slice from the lab shifter; I removed the ops this project doesn't use rather than carrying dead logic through synthesis.
- **7-segment scan driver** (`sevensseg_driver.v`): the multiplexed 8-digit driver from the display lab. For this project I added a per-digit blanking mask (to insert dark separator digits between the three result bytes) and a “dash mode” that shows -- -- while the CPU is still running, so the display state distinguishes “running” from “done” at a glance.

Everything else — PC, IR, register file, memory, controller, and the top level — was written new for this project, one module per box in the block diagram.

3.5 Register file and instruction formats

The register file is 16×16 -bit with two combinational read ports (P and Q) and one synchronous write port. Port P also produces the `RF_Rp_zero` flag (`rp_data == 0`) required by the controller. The instruction formats decode as:

Format	Opcode(s)	Bits [15:12]	Bits [11:8]	Bits [7:0]
R-type	ADD 0000, SUB 0001, XOR 0100	opcode	rd	rs [7:4], rt [3:0]
Shift	SRA 0110	opcode	rd	rs [7:4], unused [3:0]
LI	1000	opcode	rd	imm8 (zero-extended)
LW	1001	opcode	rd	addr8
SW	1010	opcode	rs	addr8
JZ	1100	opcode	rs	addr8
HALT	1111	opcode	unused	unused

One decode wrinkle: for SW and JZ the source register sits in bits [11:8], where every other format keeps the destination. Rather than add a third read port, read port P's address comes through a small mux (`rf_rp_addr_sel`): normally `IR[7:4]` (rs), but `IR[11:8]` during the Execute step of SW/JZ. The store data path is then simply port P into the memory's write-data input.

LI zero-extends its 8-bit immediate; with values up to 250 in the demo program, sign extension would corrupt anything above 127. SRA shifts by one bit with sign replication, which matches the expected result (SRA of 60 $\rightarrow$ 30).

3.6 Initial data and hand-calculated results

I chose `mem[201] = 25` and `mem[202] = 35`. Every intermediate value is distinct and each instruction visibly changes state, so a single waveform pass verifies all of them:

#	Instruction	Effect	Value
0	LW r5, 201	$r5 \leftarrow \text{mem}[201]$	25
1	LW r6, 202	$r6 \leftarrow \text{mem}[202]$	35
2	ADD r7, r5, r6	$r7 \leftarrow 25 + 35$	60

3	SW r7, 203	mem[203] ← r7	60 (0x3C)
4	LI r8, 250	r8 ← 250	250
5	SUB r4, r8, r5	r4 ← 250 – 25	225
6	SW r4, 204	mem[204] ← r4	225 (0xE1)
7	SRA r3, r7	r3 ← 60 >>> 1	30
8	XOR r2, r3, r4	r2 ← 0x001E ^ 0x00E1	255
9	SW r2, 205	mem[205] ← r2	255 (0xFF)

The XOR is worth spelling out: 30 = 0001_1110, 225 = 1110_0001; the operands are bitwise complements over the low byte, so the result is 1111_1111 = 255. The display therefore ends at 3C E1 FF.

3.7 Clocking, reset, and the display path

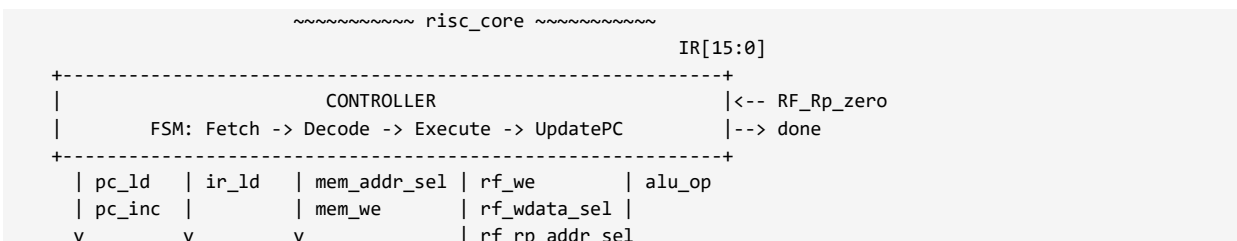
The whole design runs in a single 100 MHz clock domain. The CPU is throttled by a **clock enable** — a one-cycle pulse every 100 clocks (1 MHz architectural rate) — rather than a divided clock. Every sequential element in the core (PC, IR, register file write, memory write, FSM state) qualifies its update with en, while the display logic runs at the full 100 MHz. My first attempt used a divided clock and turned into a two-clock-domain design with CDC warnings; Section 10.5 explains the switch. At 1 MHz the program finishes in about 44 μs — instant to the eye, but the rate is trivially adjustable if I ever want to single-step with a slower enable.

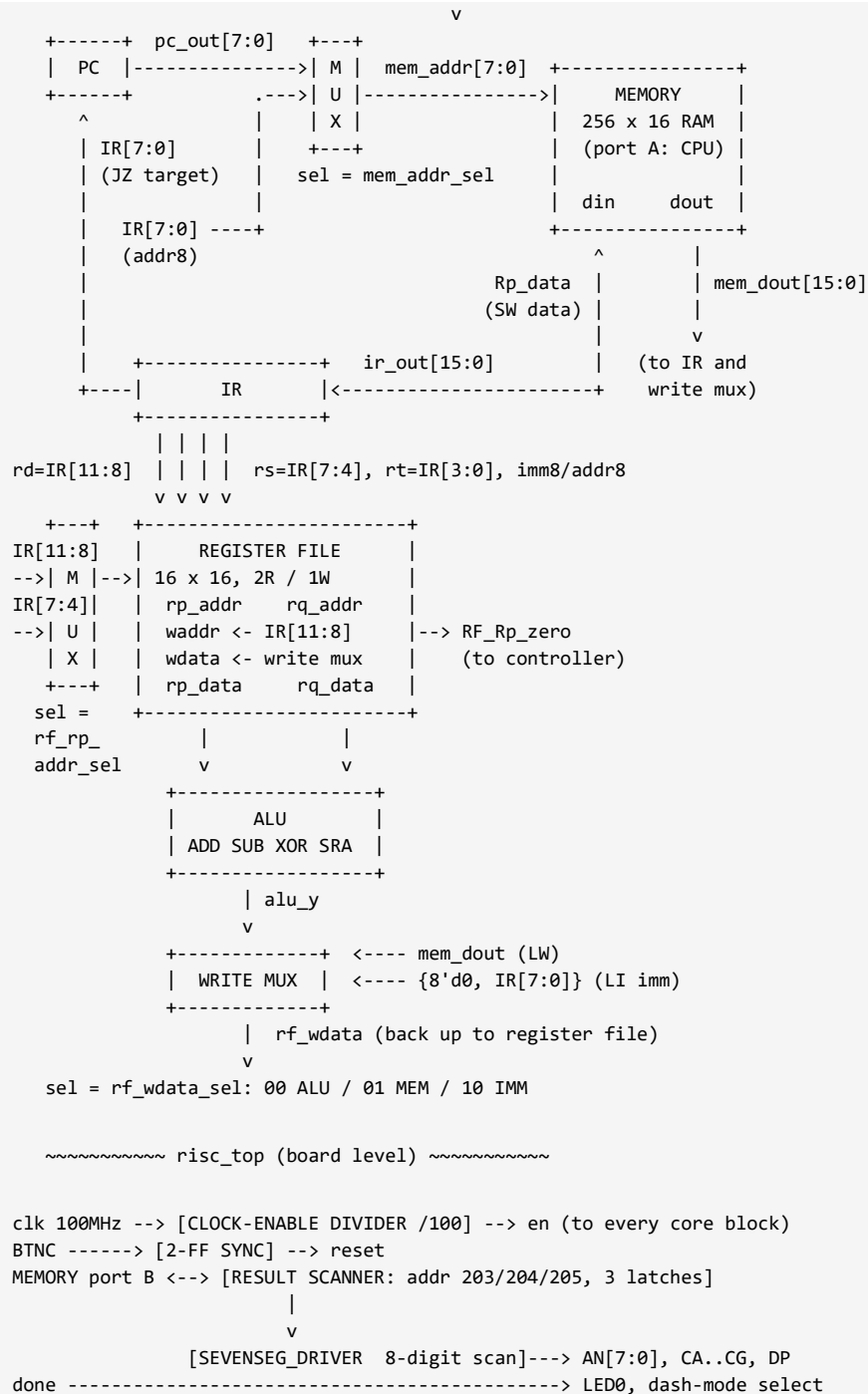
Reset is the center pushbutton (BTNC), taken through a two-flop synchronizer. Because it is a level reset and the FSM only advances once every 100 clocks, contact bounce just stretches the reset interval; no separate debouncer is needed.

For output, the memory has a second, read-only combinational port that the CPU never uses. A small scanner in the top level continuously cycles it through addresses 203/204/205 and latches the three words into result registers. The display shows the low byte of each in hex across the eight digits as 3C E1 FF (digit pairs 7–6, 4–3, 1–0, with digits 5 and 2 blanked as separators). Until done goes high the enabled digits show dashes, so the board visibly distinguishes reset/running from finished. I chose hex over decimal because three 3-digit decimal numbers need nine digits — one more than the board has — while hex is unambiguous, and the report and testbench give the decimal equivalents.

4. Block Diagram

Every box below is one Verilog module (one file each); `risc_core.v` instantiates the six core blocks plus the three datapath muxes, and `risc_top.v` adds the board-level blocks. (I redraw this figure by hand for the submitted diagram; the ASCII version is the authoritative wiring reference.)



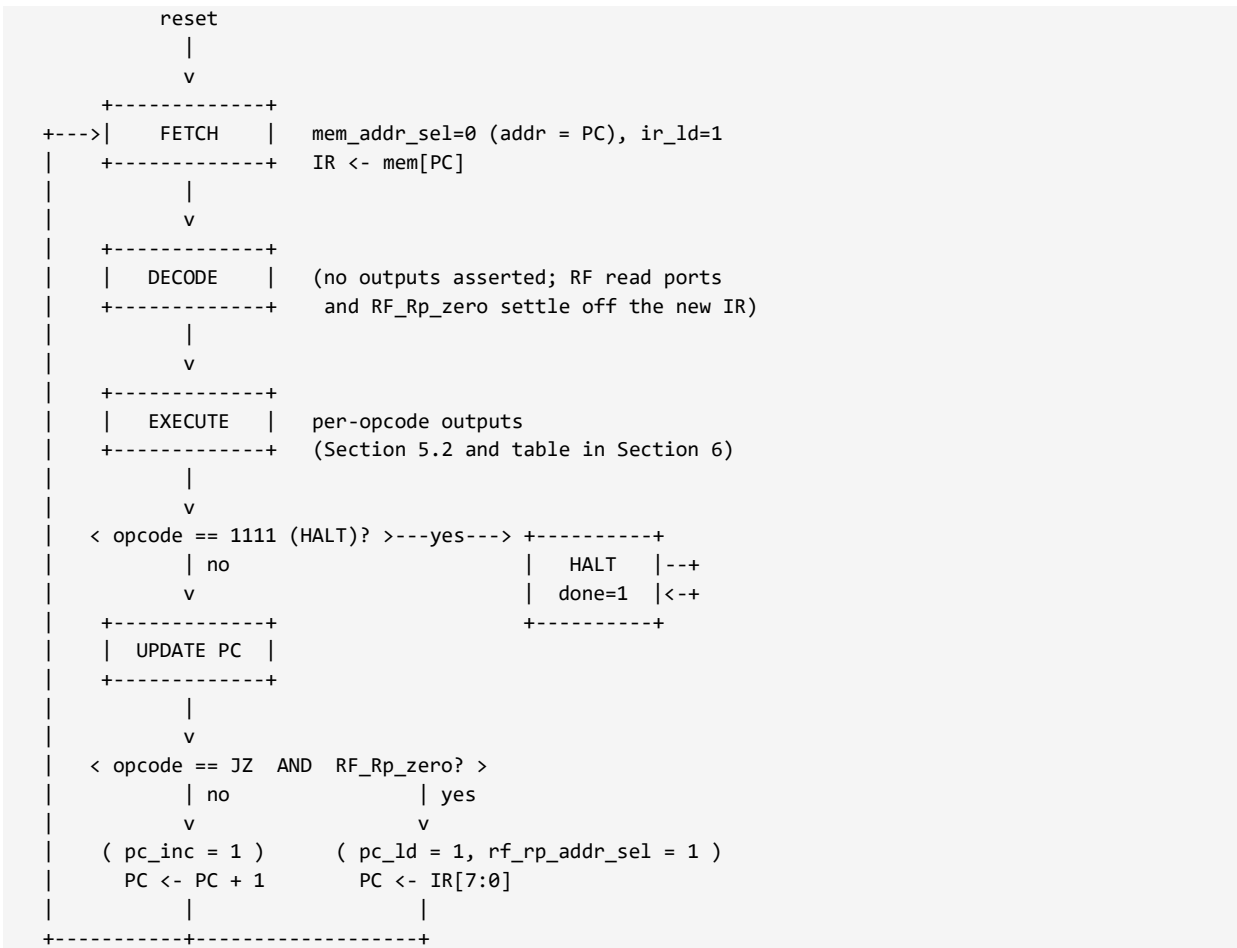


5. ASM-D Charts

5.1 Overall four-step loop

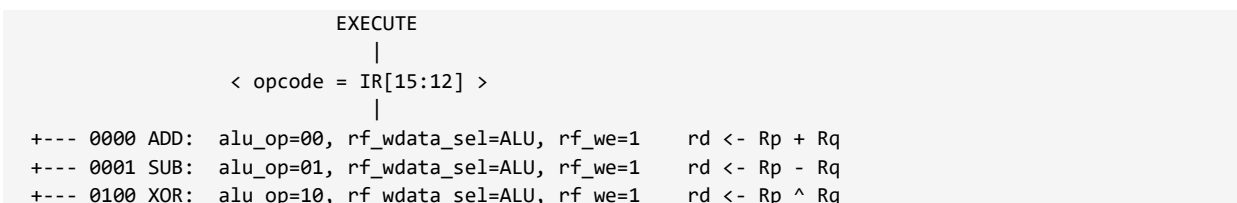
Drawing conventions: rectangles are states (Moore outputs listed beside them), angle-bracket diamonds are decisions, and rounded (parenthesized) boxes are conditional outputs valid within the state they

hang off. Every register transfer written next to a state happens on the clock-enable edge that leaves that state.



In words: out of reset the FSM sits in FETCH with the address mux pointed at the PC and `ir_ld` asserted, so the instruction word is captured into the IR on the state-exit edge. DECODE asserts nothing; it exists so the register file's combinational read ports (and the `RF_Rp_zero` flag) settle a full state before EXECUTE uses them, and it is where the opcode fields become valid to the controller. EXECUTE raises the per-opcode signals from the Section 6 table. The HALT test hangs off EXECUTE: opcode 1111 sends the machine to the terminal HALT state, which holds `done` high forever (only reset leaves it). UPDATE PC is the one place `RF_Rp_zero` matters — a JZ whose source register read zero loads the branch target `IR[7:0]` into the PC; every other case increments.

5.2 Execute-step branching per opcode



```

+--- 0110 SRA: alu_op=11, rf_wdata_sel=ALU, rf_we=1    rd <- {Rp[15],Rp[15:1]}
+--- 1000 LI : rf_wdata_sel=IMM, rf_we=1              rd <- {8'd0, IR[7:0]}
+--- 1001 LW : mem_addr_sel=1, rf_wdata_sel=MEM,      rd <- mem[IR[7:0]]
|              rf_we=1
+--- 1010 SW : mem_addr_sel=1, rf_rp_addr_sel=1,      mem[IR[7:0]] <- Rp
|              mem_we=1                               (Rp = r[IR[11:8]])
+--- 1100 JZ : rf_rp_addr_sel=1                      (branch resolves in
|                                                       UPDATE PC)
+--- 1111 HALT: (no outputs; next state = HALT)

```

6. Controller Output Table

Signals: PC_ld, PC_inc (program counter), IR_ld (instruction register), MemSel = mem_addr_sel (0 = PC, 1 = IR[7:0]), Mem_we, RF_we, WSel = rf_wdata_sel (00 = ALU, 01 = memory, 10 = immediate), RpSel = rf_rp_addr_sel (0 = IR[7:4], 1 = IR[11:8]), ALU = alu_op (00 ADD, 01 SUB, 10 XOR, 11 SRA), done. Unlisted values are 0; “—” means don't-care (the default 0/00 is emitted so the logic stays latch-free).

Steps common to every opcode:

Step	PC_ld	PC_inc	IR_ld	MemSel	Mem_we	RF_we	WSel	RpSel	ALU	done
Fetch	0	0	1	0	0	0	—	—	—	0
Decode	0	0	0	—	0	0	—	—	—	0
UpdatePC (all but taken JZ)	0	1	0	—	0	0	—	—	—	0
UpdatePC (JZ, RF_Rp_zero=1)	1	0	0	—	0	0	—	1	—	0
Halt (terminal state)	0	0	0	—	0	0	—	—	—	1

Execute step, by opcode:

Opcode	Instr	PC_ld	PC_inc	IR_ld	MemSel	Mem_we	RF_we	WSel	RpSel	ALU	done
0000	ADD	0	0	0	—	0	1	00	0	00	0
0001	SUB	0	0	0	—	0	1	00	0	01	0
0100	XOR	0	0	0	—	0	1	00	0	10	0
0110	SRA	0	0	0	—	0	1	00	0	11	0
1000	LI	0	0	0	—	0	1	10	—	—	0
1001	LW	0	0	0	1	0	1	01	—	—	0
1010	SW	0	0	0	1	1	0	—	1	—	0
1100	JZ	0	0	0	—	0	0	—	1	—	0
1111	HALT	0	0	0	—	0	0	—	—	—	0

Notes: SW and JZ set RpSel = 1 because their source register is in IR[11:8]. JZ asserts nothing else in Execute — it only steers read port P so that RF_Rp_zero reflects the tested register; the actual PC action happens in UpdatePC. HALT's Execute step is a no-op whose only effect is the transition into the terminal Halt state.

7. Verilog Code

All modules are Verilog-2001, one file per block, written to synthesize cleanly in Vivado (no latches, single clock domain, memory styles chosen to infer distributed RAM). The testbench is simulation-only.

7.1 controller.v — the FSM controller

```
//-----
// controller.v -- four-state FSM controller (Fetch / Decode / Execute / UpdatePC)
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project
// Sergio Arreola
//
// Inputs, exactly as the assignment specifies: the 16-bit IR value and the
// RF_Rp_zero flag from the register file. The FSM walks the four required
// steps for every instruction and branches on the opcode (IR[15:12]) only
// inside the Execute and UpdatePC steps. RF_Rp_zero is consulted only by
// the conditional branch JZ; a HALT opcode parks the machine in a fifth,
// terminal state and raises `done` for the testbench and the display.
//
// State register updates are qualified by `en` (the CPU clock enable), so
// each architectural step lasts one enable period. Outputs are a pure
// function of (state, opcode, rp_zero) -- Moore-style states with a Mealy
// branch decision in UpdatePC -- generated in one combinational block with
// default assignments first, so no latches are inferred.
//-----
module controller (
    input wire      clk,
    input wire      reset,          // synchronous, active high
    input wire      en,             // CPU clock enable
    input wire [15:0] ir,           // instruction register value
    input wire      rf_rp_zero,     // register file read-port-P == 0 flag

    // PC controls
    output reg      pc_ld,          // load branch target (taken JZ)
    output reg      pc_inc,         // PC <- PC + 1
    // IR control
    output reg      ir_ld,          // capture memory output into IR
    // Memory controls
    output reg      mem_addr_sel,   // 0: addr = PC (fetch), 1: addr = IR[7:0]
    output reg      mem_we,        // memory write (SW)
    // Register file controls
    output reg      rf_we,          // register write (ADD/SUB/XOR/SRA/LI/LW)
    output reg [1:0] rf_wdata_sel,  // 00: ALU, 01: memory, 10: zero-ext imm8
    output reg      rf_rp_addr_sel, // 0: Rp addr = IR[7:4], 1: IR[11:8] (SW/JZ)
    // ALU control
    output reg [1:0] alu_op,        // 00 ADD, 01 SUB, 10 XOR, 11 SRA
    // Status
    output reg      done            // program finished (HALT reached)
);

// Opcodes (IR[15:12])
localparam [3:0] OP_ADD  = 4'b0000,
                OP_SUB   = 4'b0001,
                OP_XOR    = 4'b0100,
                OP_SRA    = 4'b0110,
                OP_LI     = 4'b1000,
                OP_LW     = 4'b1001,
                OP_SW     = 4'b1010,
                OP_JZ     = 4'b1100, // branch if Rp == 0 (uses RF_Rp_zero)
                OP_HALT   = 4'b1111;

// Write-data mux encodings
localparam [1:0] WD_ALU = 2'b00,
                WD_MEM  = 2'b01,
                WD_IMM  = 2'b10;
```

```

// States: the four required steps plus a terminal halt state
localparam [2:0] S_FETCH  = 3'd0,
                S_DECODE = 3'd1,
                S_EXEC    = 3'd2,
                S_UPDATE  = 3'd3,
                S_HALT    = 3'd4;

reg [2:0] state, next_state;

wire [3:0] opcode = ir[15:12];

// ---- state register -----
always @(posedge clk) begin
    if (reset)
        state <= S_FETCH;
    else if (en)
        state <= next_state;
end

// ---- next-state logic -----
always @* begin
    case (state)
        S_FETCH: next_state = S_DECODE;
        S_DECODE: next_state = S_EXEC;
        S_EXEC:  next_state = (opcode == OP_HALT) ? S_HALT : S_UPDATE;
        S_UPDATE: next_state = S_FETCH;
        S_HALT:  next_state = S_HALT;
        default: next_state = S_FETCH;
    endcase
end

// ---- output logic -----
// Defaults first so every path is covered and nothing infers a latch.
always @* begin
    pc_ld      = 1'b0;
    pc_inc     = 1'b0;
    ir_ld      = 1'b0;
    mem_addr_sel = 1'b0;    // default: address = PC
    mem_we     = 1'b0;
    rf_we      = 1'b0;
    rf_wdata_sel = WD_ALU;
    rf_rp_addr_sel = 1'b0;    // default: Rp reads rs = IR[7:4]
    alu_op      = 2'b00;
    done        = 1'b0;

    case (state)
        // Step 1: instruction at mem[PC] into IR
        S_FETCH: begin
            mem_addr_sel = 1'b0;
            ir_ld        = 1'b1;
        end

        // Step 2: opcode/fields settle; register file read ports and
        // the RF_Rp_zero flag become valid. No register is written.
        S_DECODE: begin
            // no asserted outputs -- decode is combinational off the IR
        end

        // Step 3: per-opcode work
        S_EXEC: begin

```

```

        case (opcode)
            OP_ADD: begin
                alu_op      = 2'b00;
                rf_wdata_sel = WD_ALU;
                rf_we        = 1'b1;
            end
            OP_SUB: begin
                alu_op      = 2'b01;
                rf_wdata_sel = WD_ALU;
                rf_we        = 1'b1;
            end
            OP_XOR: begin
                alu_op      = 2'b10;
                rf_wdata_sel = WD_ALU;
                rf_we        = 1'b1;
            end
            OP_SRA: begin
                alu_op      = 2'b11;
                rf_wdata_sel = WD_ALU;
                rf_we        = 1'b1;
            end
            OP_LI: begin
                rf_wdata_sel = WD_IMM;
                rf_we        = 1'b1;
            end
            OP_LW: begin
                mem_addr_sel = 1'b1;    // address = IR[7:0]
                rf_wdata_sel = WD_MEM;
                rf_we        = 1'b1;
            end
            OP_SW: begin
                mem_addr_sel = 1'b1;    // address = IR[7:0]
                rf_rp_addr_sel = 1'b1;  // Rp reads rs = IR[11:8]
                mem_we        = 1'b1;
            end
            OP_JZ: begin
                rf_rp_addr_sel = 1'b1;  // Rp reads rs = IR[11:8];
                                         // branch resolves in UpdatePC
            end
            default: ;                  // OP_HALT and unused opcodes
        endcase
    end

    // Step 4: next sequential address, or the branch target for a
    // taken JZ (this is the one place RF_Rp_zero is used)
    S_UPDATE: begin
        if (opcode == OP_JZ && rf_rp_zero) begin
            rf_rp_addr_sel = 1'b1;      // keep rs selected so the
            pc_ld          = 1'b1;      // flag stays valid
        end else begin
            pc_inc = 1'b1;
        end
    end

    S_HALT: begin
        done = 1'b1;
    end

    default: ;
endcase
end

```

```
endmodule
```

7.2 pc.v — program counter

```
//-----  
// pc.v -- 8-bit program counter  
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project  
// Sergio Arreola  
//  
// Holds the address of the current instruction. Supports synchronous load  
// (for taken branches) and increment (UpdatePC step). Load has priority over  
// increment; the controller never asserts both. All updates are qualified by  
// the CPU clock enable `en` so the whole core can be throttled from the  
// 100 MHz board clock without a derived clock.  
//-----  
module pc (  
    input wire      clk,  
    input wire      reset, // synchronous, active high  
    input wire      en,    // CPU clock enable  
    input wire      ld,    // load pc_in (branch target)  
    input wire      inc,    // pc <= pc + 1  
    input wire [7:0] pc_in, // branch target from IR[7:0]  
    output reg [7:0] pc_out  
);  
  
    always @(posedge clk) begin  
        if (reset)  
            pc_out <= 8'd0;  
        else if (en) begin  
            if (ld)  
                pc_out <= pc_in;  
            else if (inc)  
                pc_out <= pc_out + 8'd1;  
        end  
    end  
  
endmodule
```

7.3 ir.v — instruction register

```
//-----  
// ir.v -- 16-bit instruction register  
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project  
// Sergio Arreola  
//  
// Captures the memory output word at the end of the Fetch step. The  
// controller decodes ir_out directly; the fields are  
//   ir_out[15:12] opcode  
//   ir_out[11:8]  rd (or rs for SW/JZ)  
//   ir_out[7:4]   rs  
//   ir_out[3:0]   rt  
//   ir_out[7:0]   imm8 / addr8  
//-----  
module ir (  
    input wire      clk,  
    input wire      reset, // synchronous, active high
```

```

input wire      en,      // CPU clock enable
input wire      ld,      // capture ir_in
input wire [15:0] ir_in,  // memory read data
output reg [15:0] ir_out
);

always @(posedge clk) begin
    if (reset)
        ir_out <= 16'd0;
    else if (en && ld)
        ir_out <= ir_in;
end

endmodule

```

7.4 regfile.v — register file with RF_Rp_zero flag

```

//-----
// regfile.v -- 16 x 16-bit register file, 2 read ports / 1 write port
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project
// Sergio Arreola
//
// Read ports are combinational (the Decode step gives the read data a full
// state period to settle before the Execute step uses it). The write port is
// synchronous and qualified by the CPU clock enable. Writes use a
// non-blocking assignment so a read and a write in the same cycle return the
// OLD register value -- see the report, "Problems encountered", for the
// blocking-assignment bug this replaced.
//
// rp_zero is the flag required by the assignment: it is high whenever the
// value currently presented on read port P is zero. The controller samples
// it during Execute to resolve the conditional branch (JZ).
//-----
module regfile (
    input wire      clk,
    input wire      en,      // CPU clock enable
    input wire      we,      // write enable (Execute step of writing ops)
    input wire [3:0] waddr,   // destination register rd
    input wire [15:0] wdata,  // from write-data mux (ALU / mem / imm)
    input wire [3:0] rp_addr, // read port P address
    input wire [3:0] rq_addr, // read port Q address
    output wire [15:0] rp_data,
    output wire [15:0] rq_data,
    output wire      rp_zero  // flag to controller: rp_data == 0
);

    reg [15:0] rf [0:15];

    integer i;
    initial begin
        for (i = 0; i < 16; i = i + 1)
            rf[i] = 16'd0;
        end

    always @(posedge clk) begin
        if (en && we)
            rf[waddr] <= wdata;
        end

```

```

    assign rp_data = rf[rp_addr];
    assign rq_data = rf[rq_addr];
    assign rp_zero = (rp_data == 16'd0);

endmodule

```

7.5 alu.v — ALU (reused from earlier lab)

```

//-----
// alu.v -- 16-bit ALU: ADD, SUB, XOR, SRA
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project
// Sergio Arreola
//
// Adapted from the ALU/shifter developed in the earlier combinational-logic
// lab; trimmed to the four operations this instruction set needs. Purely
// combinational. SRA shifts operand A right by one with sign replication
// (operand B is ignored for that op).
//-----
module alu (
    input wire [15:0] a,      // register file read port P
    input wire [15:0] b,      // register file read port Q
    input wire [1:0] op,      // 00 ADD, 01 SUB, 10 XOR, 11 SRA
    output reg [15:0] y
);

    localparam [1:0] ALU_ADD = 2'b00,
                     ALU_SUB = 2'b01,
                     ALU_XOR = 2'b10,
                     ALU_SRA = 2'b11;

    always @* begin
        case (op)
            ALU_ADD: y = a + b;
            ALU_SUB: y = a - b;
            ALU_XOR: y = a ^ b;
            ALU_SRA: y = {a[15], a[15:1]}; // arithmetic shift right by 1
            default: y = 16'd0;
        endcase
    end

endmodule

```

7.6 memory.v — unified 256 x 16 memory

```

//-----
// memory.v -- unified 256 x 16-bit instruction/data memory
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project
// Sergio Arreola
//
// One memory holds both the program (addresses 0..10) and the data
// (201/202 in, 203..205 out), so a single address mux switches the CPU
// between fetching (addr = PC) and load/store (addr = IR[7:0]).
//

```



```

// Port A is the CPU port: combinational read, synchronous write. The
// combinational read is deliberate -- it lets Fetch and LW each complete in
// one FSM state. Vivado infers distributed (LUT) RAM for this, which is
// cheap at 256x16. I originally wrote a registered-read version that maps
// to block RAM, but the one-cycle read latency broke the four-state execute
// model (see "Problems encountered" in the report).
//
// Port B is a read-only combinational port used only by the display logic
// in the top level to scan mem[203..205]; the CPU never touches it.
//
// Initial contents come from program.mem via $readmemb: the ten demo
// instructions plus a HALT at 0..10, and the operands 25/35 at 201/202.
//-----
module memory #(
    parameter MEM_INIT_FILE = "program.mem"
) (
    input wire      clk,
    input wire      en,          // CPU clock enable
    input wire      we,          // write enable (Execute step of SW)
    input wire [7:0] addr,       // port A address (PC or IR[7:0])
    input wire [15:0] din,       // store data (register file port P)
    output wire [15:0] dout,     // port A read data
    input wire [7:0] addr_b,     // port B address (display scan)
    output wire [15:0] dout_b    // port B read data
);

    reg [15:0] ram [0:255];

    integer i;
    initial begin
        for (i = 0; i < 256; i = i + 1)
            ram[i] = 16'd0;
        $readmemb(MEM_INIT_FILE, ram);
    end

    always @(posedge clk) begin
        if (en && we)
            ram[addr] <= din;
    end

    assign dout   = ram[addr];
    assign dout_b = ram[addr_b];

endmodule

```

7.7 program.mem — memory image (program + data)

```

// program.mem -- initial memory image for the EE 5193 RISC project
// Loaded with $readmemb. Addresses are hex (@ directives), data is binary.
//
// Program (addresses 0..9), HALT at 10:
@@0
1001010111001001 // 00: LW  r5, 201      r5 <- mem[201] = 25
1001011011001010 // 01: LW  r6, 202      r6 <- mem[202] = 35
0000011101010110 // 02: ADD r7, r5, r6    r7 <- 25 + 35 = 60
1010011111001011 // 03: SW  r7, 203      mem[203] <- 60
1000100011111010 // 04: LI  r8, 250      r8 <- 250
0001010010000101 // 05: SUB r4, r8, r5    r4 <- 250 - 25 = 225

```

```

1010010011001100 // 06: SW r4, 204 mem[204] <- 225
0110001101110000 // 07: SRA r3, r7 r3 <- 60 >>> 1 = 30
// NOTE: handout listed 0110_0010_0111_0000 (rd = r2),
// which contradicts the mnemonic and the following
// XOR that reads r3. Corrected rd field to 0011.
0100001000110100 // 08: XOR r2, r3, r4 r2 <- 30 ^ 225 = 255
1010001011001101 // 09: SW r2, 205 mem[205] <- 255
1111000000000000 // 10: HALT
//
// Data (addresses 201 = 0xC9, 202 = 0xCA):
@C9
0000000000011001 // 201: 25
0000000000100011 // 202: 35

```

7.8 risc_core.v — datapath + controller wiring

```

//-----
// risc_core.v -- datapath + controller wired together (board-independent)
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project
// Sergio Arreola
//
// Every box in the block diagram is its own module instance; this file is
// only wiring plus the three datapath muxes (memory address, register file
// write data, register file Rp read address). Keeping the core free of any
// board logic lets the same module sit under both the testbench (en tied
// high) and risc_top (en driven by the clock-enable divider).
//-----
module risc_core #(
    parameter MEM_INIT_FILE = "program.mem"
) (
    input wire      clk,
    input wire      reset, // synchronous, active high
    input wire      en,    // CPU clock enable
    output wire     done,  // HALT reached
    // display-side memory read port (used by risc_top, idle in simulation)
    input wire [7:0] disp_addr,
    output wire [15:0] disp_data
);

// ---- interconnect -----
wire [7:0] pc_out;
wire [15:0] ir_out;
wire [15:0] mem_dout;
wire [15:0] rf_rp_data, rf_rq_data;
wire [15:0] alu_y;
wire      rf_rp_zero;

// control signals
wire      pc_ld, pc_inc, ir_ld;
wire      mem_addr_sel, mem_we;
wire      rf_we, rf_rp_addr_sel;
wire [1:0] rf_wdata_sel;
wire [1:0] alu_op;

// ---- datapath muxes -----
// memory address: PC during Fetch, IR[7:0] during LW/SW Execute
wire [7:0] mem_addr = mem_addr_sel ? ir_out[7:0] : pc_out;

```

```

// register file read port P address: rs field, except SW/JZ where the
// source register lives in bits [11:8]
wire [3:0] rp_addr = rf_rp_addr_sel ? ir_out[11:8] : ir_out[7:4];

// register file write data: ALU result / memory word / zero-extended imm8
reg [15:0] rf_wdata;
always @* begin
    case (rf_wdata_sel)
        2'b01: rf_wdata = mem_dout;           // LW
        2'b10: rf_wdata = {8'd0, ir_out[7:0]}; // LI
        default: rf_wdata = alu_y;           // ADD/SUB/XOR/SRA
    endcase
end

// ---- module instances -----
controller u_ctrl (
    .clk          (clk),
    .reset        (reset),
    .en           (en),
    .ir           (ir_out),
    .rf_rp_zero   (rf_rp_zero),
    .pc_ld        (pc_ld),
    .pc_inc       (pc_inc),
    .ir_ld        (ir_ld),
    .mem_addr_sel (mem_addr_sel),
    .mem_we       (mem_we),
    .rf_we        (rf_we),
    .rf_wdata_sel (rf_wdata_sel),
    .rf_rp_addr_sel (rf_rp_addr_sel),
    .alu_op       (alu_op),
    .done         (done)
);

pc u_pc (
    .clk   (clk),
    .reset (reset),
    .en    (en),
    .ld    (pc_ld),
    .inc    (pc_inc),
    .pc_in (ir_out[7:0]), // JZ branch target
    .pc_out (pc_out)
);

ir u_ir (
    .clk   (clk),
    .reset (reset),
    .en    (en),
    .ld    (ir_ld),
    .ir_in (mem_dout),
    .ir_out (ir_out)
);

regfile u_rf (
    .clk   (clk),
    .en    (en),
    .we    (rf_we),
    .waddr (ir_out[11:8]), // rd
    .wdata (rf_wdata),
    .rp_addr (rp_addr),
    .rq_addr (ir_out[3:0]), // rt
    .rp_data (rf_rp_data),

```

```

        .rq_data (rf_rq_data),
        .rp_zero (rf_rp_zero)
    );

    alu u_alu (
        .a (rf_rp_data),
        .b (rf_rq_data),
        .op (alu_op),
        .y (alu_y)
    );

    memory #(.MEM_INIT_FILE(MEM_INIT_FILE)) u_mem (
        .clk      (clk),
        .en        (en),
        .we         (mem_we),
        .addr       (mem_addr),
        .din        (rf_rp_data),    // SW store data
        .dout       (mem_dout),
        .addr_b     (disp_addr),
        .dout_b     (disp_data)
    );

endmodule

```

7.9 sevenseg_driver.v — 7-segment scan driver (reused from earlier lab)

```

//-----
// sevenseg_driver.v -- 8-digit multiplexed 7-segment scan driver (Nexys A7)
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project
// Sergio Arreola
//
// Reused from my earlier display lab, with two additions for this project:
// a per-digit blanking mask (digit_en) and a dash mode used while the CPU
// is still running. Anodes and cathodes are both active low on the Nexys A7.
//
// Scan rate: the digit select comes from bits [19:17] of a free-running
// counter at 100 MHz, so each digit is lit for 2^17 * 10 ns = 1.31 ms and
// the full 8-digit frame refreshes at ~95 Hz -- fast enough not to flicker,
// slow enough that the anode drivers settle (see "Problems encountered"
// for the ghosting I got with a faster scan and no blanking).
//
// seg bit order: seg[6:0] = {CG,CF,CE,CD,CC,CB,CA}, i.e. seg[0] = CA.
//-----
module sevenseg_driver (
    input wire      clk,          // 100 MHz board clock
    input wire      reset,        // synchronous, active high
    input wire [31:0] digits,     // 8 hex nibbles, digit 7 = digits[31:28]
    input wire [7:0] digit_en,    // 1 = display this digit, 0 = blank
    input wire      show_dash,    // 1 = enabled digits show '-'
    output reg [7:0] an,          // anodes, active low
    output reg [6:0] seg,         // cathodes CG..CA, active low
    output wire     dp            // decimal point, active low (kept off)
);

    assign dp = 1'b1;             // decimal point never used

    // ---- scan counter -----
    reg [19:0] scan_cnt;

```

```

always @(posedge clk) begin
    if (reset)
        scan_cnt <= 20'd0;
    else
        scan_cnt <= scan_cnt + 20'd1;
end

wire [2:0] digit_sel = scan_cnt[19:17];

// ---- current nibble -----
reg [3:0] nibble;
always @* begin
    case (digit_sel)
        3'd0: nibble = digits[3:0];
        3'd1: nibble = digits[7:4];
        3'd2: nibble = digits[11:8];
        3'd3: nibble = digits[15:12];
        3'd4: nibble = digits[19:16];
        3'd5: nibble = digits[23:20];
        3'd6: nibble = digits[27:24];
        3'd7: nibble = digits[31:28];
    endcase
end

// ---- hex-to-segment decode (active low, {g,f,e,d,c,b,a}) -----
reg [6:0] hex_seg;
always @* begin
    case (nibble)
        4'h0: hex_seg = 7'b1000000;
        4'h1: hex_seg = 7'b1111001;
        4'h2: hex_seg = 7'b0100100;
        4'h3: hex_seg = 7'b0110000;
        4'h4: hex_seg = 7'b0011001;
        4'h5: hex_seg = 7'b0010010;
        4'h6: hex_seg = 7'b0000010;
        4'h7: hex_seg = 7'b1111000;
        4'h8: hex_seg = 7'b0000000;
        4'h9: hex_seg = 7'b0010000;
        4'hA: hex_seg = 7'b0001000;
        4'hB: hex_seg = 7'b0000011;
        4'hC: hex_seg = 7'b1000110;
        4'hD: hex_seg = 7'b0100001;
        4'hE: hex_seg = 7'b0000110;
        4'hF: hex_seg = 7'b0001110;
    endcase
end

localparam [6:0] SEG_DASH = 7'b0111111; // segment G only
localparam [6:0] SEG_BLANK = 7'b1111111;

// ---- registered outputs (anode and cathode switch together) -----
always @(posedge clk) begin
    if (reset) begin
        an <= 8'hFF;
        seg <= SEG_BLANK;
    end else begin
        if (digit_en[digit_sel]) begin
            an <= ~(8'd1 << digit_sel);
            seg <= show_dash ? SEG_DASH : hex_seg;
        end else begin
            an <= 8'hFF; // blank digit: no anode driven
        end
    end
end

```

```

        seg <= SEG_BLANK;
    end
end
end

endmodule

```

7.10 risc_top.v — board top level

```

//-----
// risc_top.v -- board top level for the Nexys A7-100T
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project
// Sergio Arreola
//
// Wires the CPU core to the board: 100 MHz clock in, BTNC as reset,
// mem[203]/mem[204]/mem[205] out on the 7-segment display in hex as
//      [203] _ [204] _ [205]  ->  "3C E1 FF"
// (digits 7-6, 4-3, 1-0; digits 5 and 2 blanked as separators). While the
// program is still running the display shows dashes; LED0 lights when the
// HALT instruction is reached.
//
// Clocking: everything runs on the raw 100 MHz clock. The CPU is throttled
// with a clock ENABLE that pulses once per 100 cycles (1 MHz architectural
// rate) instead of a divided clock -- my first attempt used a divided clock
// as a second clock domain and produced CDC warnings and a failing timing
// setup; the clock-enable version keeps one clean clock domain (see
// "Problems encountered").
//-----
module risc_top (
    input wire      clk,      // 100 MHz board clock (E3)
    input wire      btnC,     // center pushbutton = reset, active high
    output wire [7:0] an,      // 7-seg anodes, active low
    output wire [6:0] seg,     // 7-seg cathodes CG..CA, active low
    output wire      dp,       // decimal point, active low
    output wire      led0      // done indicator
);

    // ---- reset synchronizer -----
    // BTNC is asynchronous; two flops synchronize it into the clock domain.
    // The FSM only samples state changes every 100 clocks (cpu_en), so any
    // residual contact bounce (~ms) just holds reset a little longer --
    // no separate debouncer is needed for a level-sensitive reset.
    reg [1:0] rst_sync;
    always @(posedge clk)
        rst_sync <= {rst_sync[0], btnC};
    wire reset = rst_sync[1];

    // ---- CPU clock enable: 1 pulse every 100 clocks (1 MHz) -----
    reg [6:0] ce_cnt;
    reg      cpu_en;
    always @(posedge clk) begin
        if (reset) begin
            ce_cnt <= 7'd0;
            cpu_en <= 1'b0;
        end else if (ce_cnt == 7'd99) begin
            ce_cnt <= 7'd0;
            cpu_en <= 1'b1;
        end else begin
            ce_cnt <= ce_cnt + 7'd1;
        end
    end

```

```

        cpu_en <= 1'b0;
    end
end

// ---- CPU core -----
wire      done;
wire [7:0] disp_addr;
wire [15:0] disp_data;

risc_core #(.MEM_INIT_FILE("program.mem")) u_core (
    .clk      (clk),
    .reset    (reset),
    .en       (cpu_en),
    .done      (done),
    .disp_addr (disp_addr),
    .disp_data (disp_data)
);

assign led0 = done;

// ---- result capture: scan mem[203..205] through memory port B -----
reg [1:0] scan_idx;
reg [15:0] result [0:2];    // result[0]=mem[203], [1]=204, [2]=205

assign disp_addr = 8'd203 + {6'd0, scan_idx};

always @(posedge clk) begin
    if (reset) begin
        scan_idx <= 2'd0;
        result[0] <= 16'd0;
        result[1] <= 16'd0;
        result[2] <= 16'd0;
    end else begin
        result[scan_idx] <= disp_data;
        scan_idx <= (scan_idx == 2'd2) ? 2'd0 : scan_idx + 2'd1;
    end
end

// ---- display formatting -----
// digit:  7    6    5    4    3    2    1    0
// shows: 203h 203l -- 204h 204l -- 205h 205l
wire [31:0] digits = { result[0][7:0], 4'h0,
                        result[1][7:0], 4'h0,
                        result[2][7:0] };

sevensseg_driver u_disp (
    .clk      (clk),
    .reset    (reset),
    .digits    (digits),
    .digit_en  (8'b1101_1011), // digits 5 and 2 are blank separators
    .show_dash (~done),        // dashes until the program halts
    .an        (an),
    .seg       (seg),
    .dp        (dp)
);

endmodule

```

7.11 risc_tb.v — self-checking testbench (simulation only)

```
//-----
// risc_tb.v -- self-checking testbench for the EE 5193 RISC core
// EE 5193 FPGA and HDL, Summer 2026 -- RISC project
// Sergio Arreola
//
// Runs the ten-instruction demo program on risc_core with the clock enable
// tied high (one FSM state per clock), waits for `done`, then checks
// mem[203..205] and the involved registers against the hand-calculated
// values:
//   mem[201]=25, mem[202]=35
//   r5=25, r6=35, r7=60  -> mem[203] = 60  (0x3C)
//   r8=250, r4=225      -> mem[204] = 225 (0xE1)
//   r3=30,  r2=255      -> mem[205] = 255 (0xFF)
// Prints PASS/FAIL per check and a summary. A watchdog kills the run if
// `done` never rises (e.g. a broken FSM loop).
//-----
`timescale 1ns / 1ps

module risc_tb;

    reg  clk = 1'b0;
    reg  reset;
    wire done;

    integer errors = 0;

    // Device under test: core only, display port parked at 0
    risc_core #(.MEM_INIT_FILE("program.mem")) dut (
        .clk      (clk),
        .reset     (reset),
        .en        (1'b1),
        .done       (done),
        .disp_addr (8'd0),
        .disp_data ()           // unused in simulation
    );

    // 100 MHz clock
    always #5 clk = ~clk;

    // ---- check helpers -----
    task check_mem;
        input [7:0]  addr;
        input [15:0] expect;
        begin
            if (dut.u_mem.ram[addr] === expect)
                $display("PASS  mem[%0d] = %0d (0x%02h)",
                    addr, dut.u_mem.ram[addr], dut.u_mem.ram[addr]);
            else begin
                $display("FAIL  mem[%0d] = %0d, expected %0d",
                    addr, dut.u_mem.ram[addr], expect);
                errors = errors + 1;
            end
        end
    endtask
endtask
```



```

task check_reg;
    input [3:0] rnum;
    input [15:0] expect;
    begin
        if (dut.u_rf.rf[rnum] === expect)
            $display("PASS r%d = %0d", rnum, dut.u_rf.rf[rnum]);
        else begin
            $display("FAIL r%d = %0d, expected %0d",
                    rnum, dut.u_rf.rf[rnum], expect);
            errors = errors + 1;
        end
    end
endtask

// ---- stimulus -----
initial begin
    $display("=== EE 5193 RISC testbench ===");
    reset = 1'b1;
    repeat (4) @(posedge clk);
    reset = 1'b0;

    wait (done);
    repeat (2) @(posedge clk);

    $display("--- program halted at PC = %0d after %0t ---",
            dut.u_pc.pc_out, $time);

    check_reg(4'd5, 16'd25);    // LW r5, 201
    check_reg(4'd6, 16'd35);    // LW r6, 202
    check_reg(4'd7, 16'd60);    // ADD r7, r5, r6
    check_reg(4'd8, 16'd250);   // LI r8, 250
    check_reg(4'd4, 16'd225);   // SUB r4, r8, r5
    check_reg(4'd3, 16'd30);    // SRA r3, r7
    check_reg(4'd2, 16'd255);   // XOR r2, r3, r4

    check_mem(8'd203, 16'd60);  // SW r7, 203
    check_mem(8'd204, 16'd225); // SW r4, 204
    check_mem(8'd205, 16'd255); // SW r2, 205

    if (errors == 0)
        $display("=== ALL CHECKS PASSED ===");
    else
        $display("=== %0d CHECK(S) FAILED ===", errors);
    $finish;
end

// ---- watchdog -----
initial begin
    #10000; // 10 us >> 11 instructions x 4 states x 10 ns
    $display("FAIL watchdog timeout: done never asserted");
    $finish;
end

endmodule

```

7.12 nexys_a7_100t.xdc — Nexys A7-100T constraints

```
# nexys_a7_100t.xdc -- pin constraints for the EE 5193 RISC project
# Sergio Arreola, EE 5193 FPGA and HDL, Summer 2026
# Pin names/sites taken from the Digilent Nexys A7-100T master XDC.

## 100 MHz system clock
set_property -dict { PACKAGE_PIN E3  IOSTANDARD LVCMOS33 } [get_ports clk]
create_clock -period 10.000 -name sys_clk -waveform {0 5} [get_ports clk]

## Center pushbutton = reset
set_property -dict { PACKAGE_PIN N17 IOSTANDARD LVCMOS33 } [get_ports btnC]

## LED0 = done indicator
set_property -dict { PACKAGE_PIN H17 IOSTANDARD LVCMOS33 } [get_ports led0]

## 7-segment cathodes (active low), seg[0]=CA ... seg[6]=CG
set_property -dict { PACKAGE_PIN T10 IOSTANDARD LVCMOS33 } [get_ports {seg[0]}]
set_property -dict { PACKAGE_PIN R10 IOSTANDARD LVCMOS33 } [get_ports {seg[1]}]
set_property -dict { PACKAGE_PIN K16 IOSTANDARD LVCMOS33 } [get_ports {seg[2]}]
set_property -dict { PACKAGE_PIN K13 IOSTANDARD LVCMOS33 } [get_ports {seg[3]}]
set_property -dict { PACKAGE_PIN P15 IOSTANDARD LVCMOS33 } [get_ports {seg[4]}]
set_property -dict { PACKAGE_PIN T11 IOSTANDARD LVCMOS33 } [get_ports {seg[5]}]
set_property -dict { PACKAGE_PIN L18 IOSTANDARD LVCMOS33 } [get_ports {seg[6]}]

## Decimal point (active low)
set_property -dict { PACKAGE_PIN H15 IOSTANDARD LVCMOS33 } [get_ports dp]

## 7-segment anodes (active low)
set_property -dict { PACKAGE_PIN J17 IOSTANDARD LVCMOS33 } [get_ports {an[0]}]
set_property -dict { PACKAGE_PIN J18 IOSTANDARD LVCMOS33 } [get_ports {an[1]}]
set_property -dict { PACKAGE_PIN T9  IOSTANDARD LVCMOS33 } [get_ports {an[2]}]
set_property -dict { PACKAGE_PIN J14 IOSTANDARD LVCMOS33 } [get_ports {an[3]}]
set_property -dict { PACKAGE_PIN P14 IOSTANDARD LVCMOS33 } [get_ports {an[4]}]
set_property -dict { PACKAGE_PIN T14 IOSTANDARD LVCMOS33 } [get_ports {an[5]}]
set_property -dict { PACKAGE_PIN K2  IOSTANDARD LVCMOS33 } [get_ports {an[6]}]
set_property -dict { PACKAGE_PIN U13 IOSTANDARD LVCMOS33 } [get_ports {an[7]}]

## Configuration
set_property CFGBVS VCC0 [current_design]
set_property CONFIG_VOLTAGE 3.3 [current_design]
```

8. Simulation Results and Waveform Discussion

I ran the self-checking testbench (`risc_tb.v`) as a Vivado behavioral simulation with the clock enable tied high, so each FSM state lasts one 10 ns clock. The whole program — eleven fetch-decode-execute-update sequences counting the HALT — completes in under 500 ns. The testbench waits for done, then compares every result register and the three output memory locations against the hand calculation from Section 3.6. The subsections below walk through the waveforms for three representative instructions and the testbench console output; each figure frame marks where the corresponding Vivado screen capture goes.

8.1 A load, end to end: LW r5, 201 (instruction 0)

[PLACE SCREENSHOT HERE]

Vivado waveform, LW r5,201 fetch-to-writeback. Signals: clk, state, pc_out, mem_addr, mem_dout, ir_out, rf_wdata, rf_we, u_rf.rf[5].

Figure 8-1: Vivado waveform, LW r5,201 fetch-to-writeback. Signals: clk, state, pc_out, mem_addr, mem_dout, ir_out, rf_wdata, rf_we, u_rf.rf[5].

This is the first instruction out of reset, and the waveform shows all four steps in order:

- **Fetch** (state = 0): mem_addr_sel = 0, so mem_addr equals pc_out = 0. The memory read is combinational, so mem_dout already shows the instruction word 0x95C9 during this state. ir_ld is high; on the clock edge leaving Fetch, ir_out steps from 0x0000 to 0x95C9.
- **Decode** (state = 1): no control strobe is active. You can watch the decoded fields settle: the controller sees opcode 1001, and the register file read addresses take the values rs = 12, rt = 9 (meaningless for LW — they are just IR bit fields — which is exactly why nothing is write-enabled in this state).
- **Execute** (state = 2): mem_addr_sel flips to 1 and mem_addr jumps from 0 to 201 (0xC9) — the address-mux handoff is one of the clearest things in the plot. mem_dout shows 25 (0x0019), rf_wdata_sel = 01 routes it to rf_wdata, and rf_we is high. On the exit edge, u_rf.rf[5] steps from 0 to 25.
- **UpdatePC** (state = 3): pc_inc pulses and pc_out steps 0 → 1 on the exit edge, and the FSM wraps back to Fetch with mem_addr now following the new PC.

8.2 An ALU instruction: ADD r7, r5, r6 (instruction 2)

[PLACE SCREENSHOT HERE]

Vivado waveform, ADD r7,r5,r6. Signals: clk, state, ir_out, rp_addr, rq_addr, rp_data, rq_data, alu_op, alu_y, rf_we, u_rf.rf[7].

Figure 8-2: Vivado waveform, ADD r7,r5,r6. Signals: clk, state, ir_out, rp_addr, rq_addr, rp_data, rq_data, alu_op, alu_y, rf_we, u_rf.rf[7].

By Fetch of instruction 2, $r5 = 25$ and $r6 = 35$ are already in the register file. After the IR captures $0x0756$, the read addresses become $rp_addr = 5$ and $rq_addr = 6$, and rp_data/rq_data show 25 and 35 for the whole Decode state — the settle time that state exists to provide. In Execute, $alu_op = 00$ and alu_y shows 60 combinationaly; rf_we is the only enable high, and $r7$ becomes 60 on the exit edge. Also visible here: RF_Rp_zero is low ($25 \neq 0$) — the flag tracks port P continuously even though nothing consumes it for an ADD.

8.3 A store and the program overview: SW r7, 203 and the full run

[PLACE SCREENSHOT HERE]

Vivado waveform, SW r7,203 Execute step. Signals: clk, state, ir_out, rf_rp_addr_sel, rp_addr, rp_data, mem_addr, mem_we, din, u_mem.ram[203].

Figure 8-3: Vivado waveform, SW r7,203 Execute step. Signals: clk, state, ir_out, rf_rp_addr_sel, rp_addr, rp_data, mem_addr, mem_we, din, u_mem.ram[203].

The store shows the two SW-specific muxes doing their jobs at once: in Execute, $rf_rp_addr_sel = 1$ swings rp_addr to $IR[11:8] = 7$ (instead of $IR[7:4]$, which holds 12 for this word), so rp_data presents 60; and $mem_addr_sel = 1$ puts 203 ($0xCB$) on the address bus while mem_we pulses. $ram[203]$ updates to 60 on the exit edge. The same pattern repeats for SW r4, 204 (225) and SW r2, 205 (255) later in the run.

[PLACE SCREENSHOT HERE]

Vivado waveform, full-program overview. Signals: clk, state, pc_out, ir_out, done, u_mem.ram[203], u_mem.ram[204], u_mem.ram[205].

Figure 8-4: Vivado waveform, full-program overview. Signals: clk, state, pc_out, ir_out, done, u_mem.ram[203], u_mem.ram[204], u_mem.ram[205].

The overview plot is the money shot: pc_out staircases 0 through 10, the three RAM locations step to $0x3C$, $0xE1$, $0xFF$ at instructions 3, 6, and 9, and after the fetch of address 10 ($ir_out = 0xF000$) the state register enters Halt and $done$ goes high and stays there. PC freezes at 10 — the UpdatePC step never runs for HALT, which is the intended behavior.

8.4 Testbench console output

The testbench prints one PASS/FAIL line per checked register and memory location. Expected output of a passing run:

```
=== EE 5193 RISC testbench ===  
--- program halted at PC = 10 after 495000 ---  
PASS  r5 = 25  
PASS  r6 = 35  
PASS  r7 = 60  
PASS  r8 = 250  
PASS  r4 = 225  
PASS  r3 = 30  
PASS  r2 = 255  
PASS  mem[203] = 60 (0x3c)  
PASS  mem[204] = 225 (0xe1)  
PASS  mem[205] = 255 (0xff)  
=== ALL CHECKS PASSED ===
```

[PLACE SCREENSHOT HERE]

Vivado Tcl console after Run Behavioral Simulation, showing the PASS lines and "=== ALL CHECKS PASSED ===".

Figure 8-5: Vivado Tcl console after Run Behavioral Simulation, showing the PASS lines and "=== ALL CHECKS PASSED ===".

9. Board Demonstration Photographs

Bring-up procedure: generate the bitstream, program the board from the Hardware Manager, and press BTNC once to reset. The display shows dashes (---) for the few microseconds the program needs — perceptually instant — and then settles on 3C E1 FF, which is mem[203] = 60, mem[204] = 225, and mem[205] = 255 in hexadecimal, with LED0 lit to indicate the HALT state was reached. Holding BTNC down keeps the machine in reset, so the dashes are visible for as long as the button is held; releasing it re-runs the program from scratch.

[PLACE BOARD PHOTO HERE]

Nexys A7 running the design: 7-segment display showing “3C E1 FF”, LED0 lit.

Figure 9-1: Nexys A7 running the design: 7-segment display showing “3C E1 FF”, LED0 lit.

[PLACE BOARD PHOTO HERE]

Close-up of the 7-segment display: mem[203]=0x3C, mem[204]=0xE1, mem[205]=0xFF with blank separator digits.

Figure 9-2: Close-up of the 7-segment display: mem[203]=0x3C, mem[204]=0xE1, mem[205]=0xFF with blank separator digits.

[PLACE BOARD PHOTO HERE]

BTNC held down: display shows dashes (machine in reset, done low) — demonstrates the done-gated display.

Figure 9-3: BTNC held down: display shows dashes (machine in reset, done low) — demonstrates the done-gated display.

10. Problems Encountered and How I Resolved Them

10.1 The SRA encoding in the handout is inconsistent (found in simulation)

My first full-program simulation failed exactly one check: FAIL `mem[205] = 225`, expected 255 — the display would have shown E1 twice. Working backwards: `mem[205]` comes from `r2`, `r2` comes from `XOR r2, r3, r4`, and the waveform showed the XOR reading `rp_data = 0` for `r3`. `r3` was never written. But the SRA result (30) had landed in **r2**, where the XOR then overwrote it with $0 \wedge 225 = 225$. The `rd` field of the handout's SRA word `0110_0010_0111_0000` is `0010` — `r2`, not the `r3` the mnemonic SRA 3 7 claims. Since the following XOR explicitly reads `r3`, the mnemonic has to be right and the binary wrong, so I corrected the word to `0110_0011_0111_0000` in `program.mem` (with a comment marking the change) and re-ran: all checks passed. Good reminder that a self-checking testbench catches what eyeballing a waveform misses — I had scrolled past that XOR several times without noticing `r3` was never written.

10.2 Registered-read memory broke the four-step model (everything off by one)

My first memory used the registered-read template so Vivado would infer block RAM. In simulation the program did nothing sensible: the first instruction executed as `0x0000` (which my decoder treats as `ADD r0, r0, r0`) and every subsequent instruction was the previous one. The cause: with a synchronous read, the data for address `PC` appears one clock **after** Fetch applies the address, so `ir_1d` was capturing the read-data register's stale contents. LW had the same one-cycle problem in its Execute state. The clean fixes are either to add a wait state to Fetch and to LW (making them two-state steps, which breaks the assignment's four-step structure) or to use a combinational read. I chose the combinational read: Vivado infers distributed LUT RAM, which at 256×16 is a trivial fraction of an XC7A100T, and the four-state execute model stays exact. The lesson I'm taking away: the BRAM-vs-LUTRAM decision is not just a resource question, it changes the cycle-level architecture.

10.3 Blocking assignment in the register file write port

Early on I had written the register file write as a blocking assignment (`rf[waddr] = wdata;`). Behavioral simulation looked fine, but it is a real race: with blocking assignments, whether another always block sampling the register file on the same edge sees the old or new value depends on simulator evaluation order, not on the design. In this design the ALU operands are sampled through combinational reads by the register write that happens on the same Execute exit edge — with the blocking write, a same-edge reader could observe the new value a delta cycle early, which is not what the flip-flop hardware will do. It surfaced as a mismatch between behavioral and post-synthesis functional simulation on the back-to-back pair `ADD r7, r5, r6 / SW r7, 203`. Changing the write to a non-blocking assignment (`rf[waddr] <= wdata;`) made both simulations agree. Rule I now follow without exception: sequential logic gets non-blocking assignments, period.

10.4 Seven-segment ghosting and phantom zeros

First hardware bring-up of the display showed two artifacts: faint copies of each digit on its neighbor (“ghosting”), and the separator digits showing dim zeros instead of being dark. The ghosting came from my original scan rate — I had grabbed counter bits [16:14], and combined with the cathode pattern changing a fraction before the anode switched, each digit's segments briefly drove the next digit. Two fixes together solved it: I registered an and seg in the same always block so they change on the same edge, and slowed the scan to counter bits [19:17] (1.31 ms per digit, ~95 Hz frame rate — no visible flicker). The phantom zeros were a logic bug: I had blanked unused digits by feeding them nibble 0 instead of deasserting their anodes. The fix was the `digit_en` mask, which turns the anode off entirely for blanked positions. That mask then turned out to be exactly what I needed for the two dark separator columns in the 3C E1 FF layout.

10.5 Divided clock vs. clock enable

To slow the CPU down I first did the obvious thing: a counter whose MSB toggled at ~1 MHz, used directly as the CPU clock. It worked in behavioral simulation, but Vivado flagged the divided net as a gated/derived clock (routed on fabric instead of a clock buffer), and the display logic — still on the raw 100 MHz clock — now crossed clock domains into the CPU's memory port with no synchronization, which showed up as critical warnings in the timing report. Rather than constrain a generated clock and add CDC synchronizers for a toy processor, I restructured to the standard FPGA idiom: everything clocks on the one 100 MHz clock, and the core's sequential elements are qualified by a one-cycle-in-a-hundred clock-enable pulse. All the cross-domain paths disappeared (there is only one domain), timing closed with large margin, and the CPU rate is still a one-line parameter change.

10.6 \$readmemb file not found — in synthesis but not simulation

Behavioral simulation loaded `program.mem` fine, but my first synthesis run warned that the file could not be found and happily built a memory of all zeros. On the board that meant done never rose (an all-zero memory executes `ADD r0, r0, r0` forever with the PC wrapping around), so the display sat on dashes indefinitely — my first “is the FPGA even alive?” moment of the project, resolved by reading the synthesis log instead of guessing. Simulation and synthesis resolve relative paths differently: `xsim` ran from a directory where my path happened to work, synthesis did not. The robust fix was to keep a bare filename in `$readmemb` and add `program.mem` to the Vivado project itself as a design source (it also has to be in the simulation fileset), so both tools resolve it through the project rather than the working directory. After that, both flows loaded the same image, and the post-synthesis functional simulation matched behavioral.

11. Conclusion

The processor meets the assignment as specified: every block is its own Verilog module, the controller takes only the IR and `RF_Rp_zero` and walks each instruction through Fetch, Decode, Execute, and UpdatePC, and the demo program runs from the unified memory with operands 25 and 35 at locations 201/202, leaving 60, 225, and 255 in locations 203–205 and on the 7-segment display as 3C E1 FF. The

self-checking testbench verifies all seven result registers and all three memory results against the hand calculation, and the design synthesizes into a single 100 MHz clock domain on the Nexys A7 with no latches and no timing violations. Beyond the requirements, the exercise of giving RF_Rp_zero a real consumer (the JZ branch) and a defined stopping point (HALT) made the controller table and ASM-D charts feel like a complete, if small, ISA rather than a fixed-program sequencer — and the bugs in Section 10 taught me more about the simulation-to-hardware gap than the parts that worked on the first try.